

Acute Respiratory Distress



Section I: Scenario Demographics

Scenario Title:	Respiratory Distress secondary to pulmonary edema		
Date of Development:	10/06/2015		
Target Learning Group:	☒ Juniors (PGY 1 - 2)	☐ Seniors (PGY $\geq$ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Lindsey McMurray
Affiliations/Institution(s):	University of Toronto
Contact E-mail (optional):	mcmurray.lindsey@gmail.com

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	1. To demonstrate basic management principles and consider differential of a patient with undifferentiated dyspnea. 2. To review basic management principles of a patient with pulmonary edema.
CRM Objectives:	1. To prioritize management steps in an unstable patient. 2. To delegate tasks as necessary and communicate clearly with team members. 3. To recognize the need to call for help.
Medical Objectives:	To demonstrate the emergent management of acute respiratory distress.

Case Summary: Brief Summary of Case Progression and Major Events
A 78 year old woman post-op from a TAH+ BSO for ovarian CA has just been transferred to the ward when she develops acute shortness of breath. When the resident arrives, the patient is in significant respiratory distress saturating 80% on RA. Oxygen and medical therapy will not adequately relieve the patient's distress. The resident will need to recognize that the patient has a Grade 3-4 LV and received 2L of fluid intra-operatively. When BiPAP is called for, it will be unavailable. Ultimately, the patient will require intubation.

References
Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). <i>Rosen's emergency medicine: Concepts and clinical practice</i> . St. Louis: Mosby.

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Section IV: Scenario Script

A. Clinical Vignette: To Read Aloud at Beginning of Case

You are on the GYNE service and have been paged by the ward nurse to attend to a 78 year old woman who is having trouble breathing. She is POD #0 from a 4 hour TAH+BSO operation for ovarian CA. She just got to the ward about 1 hour ago. You enter the patient's room she is hooked up to an IV with NS running at 150cc/hr.

B. Scenario Cast & Realism

Patient:	☐ Computerized Mannequin	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual
	☒ Mannequin		☐ Physical
	☐ Standardized Patient		☐ Emotional/Experiential
	☐ Hybrid		☐ Other:
	☐ Task Trainer		☐ N/A

Confederates Brief Description of Role

Bedside nurse	Provides additional information regarding clinical course, past medical history, chart details, latest laboratory investigations

C. Required Monitors

☒ EKG Leads/Wires	☒ Temperature Probe	☐ Central Venous Line
☒ NIBP Cuff	☒ Defibrillator Pads	☐ Capnography
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:

D. Required Equipment

☒ Gloves	☒ Nasal Prongs	☐ Scalpel
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit
☒ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit
☐ Blood Products	☒ ET Tubes	☐ Other:
☐ Intraosseous Set-up	☐ LMA	☐ Other:

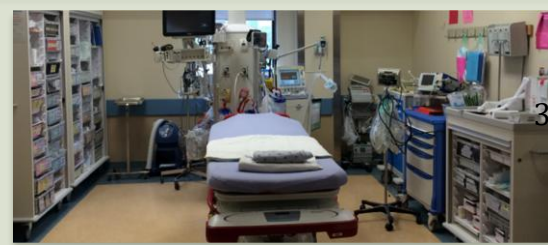
E. Moulage

Clean dressing to laparotomy scar, wound clean and dry. Diaphoresis on the forehead (spray bottle). Mock chart with admission note (for resident to decipher past medical history).

F. Approximate Timing

Set-Up:	5 min	Scenario:	10 min	Debriefing:	5 min
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Section V: Patient Data and Baseline State

A. Patient Profile and History					
Patient Name: Grace Smith		Age: 78		Weight: 80 kg	
Gender: ☐ M ☒ F		Code Status: Full			
Chief Complaint: Shortness of breath.					
History of Presenting Illness: "I can't breathe". Patient is fully alert and conscious but in respiratory distress. Nurse notes that patient received 2L of fluid in the OR.					
Past Medical History:	STEMI 2010		Medications:	Atorvastatin, Ramipril	
	CHF with grade 3-4 LV			Lasix	
	COPD			Ventolin PRN	
	HTN			ASA (on hold)	
Allergies: None					
Social History: 60 pack-year smoking, occasional alcohol, no illicit drugs					
Family History: Non contributory					
Review of Systems:	CNS:	No complaints.			
	HEENT:	No complaints.			
	CVS:	No chest pain.			
	RESP:	Sudden onset shortness of breath upon transfer to ward.			
	GI:	No complaints.			
	GU:	No complaints.			
	MSK:	No complaints.		INT:	Feels sweaty.
B. Baseline Simulator State and Physical Exam					
☒ No Monitor Display		☐ Monitor On, no data displayed		☐ Monitor on Standard Display	
HR: 110/min		BP: 150/95		RR: 34/min	
O ₂ SAT: 80%		Glucose: 6.2 mmol/L		GCS: 15	
Rhythm: NSR		T: 36.9°C			
General Status: Respiratory distress, diaphoretic, alert and following commands.					
CNS:	GCS 15.				
HEENT:	Normal				
CVS:	Pulse present, normal heart sounds.				
RESP:	Coarse crackles				
ABDO:	Dry dressing to abdomen				
GU:	Normal				
MSK:	Normal, no calf swelling.		SKIN:	Clammy, moist	

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Section VI: Scenario Progression

Scenario States, Modifiers and Triggers			
Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: NSR HR: 110/min BP: 150/95 RR: 34/min O ₂ SAT: 80% RA T: 36.9°C	Respiratory distress with significantly increased WOB but alert.	<u>Learner Actions</u> ☐ Monitors/Full vitals ☐ Supplemental O₂ ☐ Call for crash cart ☐ Call for RT, rapid response team ☐ Review patient history ☐ Focused physical exam ☐ Calls for portable CXR ☐ Trial ventolin+atrovent ☐ Send blood work (troponin, VBG, BNP) ☐ Trial Nitro sprays +/- lasix ☐ EKG ☐ Consider trial of BiPAP	<u>Modifiers</u> - NRB applied → O₂SAT to 88% - If BiPAP requested, RT says: "it'll be 10 minutes. Someone is getting it." <u>Triggers</u> - 5 min → 2. Patient Tires
2. Patient Tires HR → 120/ min RR → 24 O ₂ SAT → 85% on NRB GCS → Now drowsy	Respiratory distress continues but patient begins to tire. Patient becomes drowsy.	<u>Learner Actions</u> ☐ Repeat physical exam ☐ Consider nitro infusion ☐ Prepare for intubation ☐ Consider fentanyl pre-tx ☐ Use vaso-neutral induction ☐ Adjuncts nearby ☐ Call for help (anesthesia, ICU) ☐ Intubates patient	<u>Modifiers</u> - CXR available – show to learner as state begins - If no move toward intubation by 8 min → O₂SAT to 82% and patient unresponsive - BVM assistance → O₂SAT to 89% - BVM with PEEP → O₂SAT to 92% <u>Triggers</u> - Intubation → 3. Peri-Intubation - If BVM with PEEP and choose to wait for help → END CASE
3. Peri-Intubation HR → 90 BP → 155/85 RR → 12 (vent) O ₂ SAT → 82% with intubation then to 94% after 45 sec	Unchanged.	<u>Learner Actions</u> ☐ Start nitro infusion ☐ Post-intubation CXR ☐ Place OG ☐ Post-intubation sedation ☐ Call ICU ☐ Reassess patient, consider other diagnoses	<u>END CASE PRN</u>

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Section VII: Supporting Documents, Laboratory Results, & Multimedia

Images (ECGs, CXRs, etc.)

ECG

<https://thejarvik7.files.wordpress.com/2012/02/inferior-wall-stemi-2005-05-27-08.jpg>

CXR

<https://www.med-ed.virginia.edu/courses/rad/cxr/pathology2chest.html>

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Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
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Medical Objectives:	To demonstrate the emergent management of acute respiratory distress.
Sample Questions for Debriefing	
1. What do you think your team did really well with this critically ill patient? 2. What were some communication difficulties the team experienced? 3. What are your top priorities on arriving to assess a patient in respiratory distress? 4. What are your immediate options to improve patient's state? 5. Who can you call for help? 6. What else is on the differential for respiratory distress? 7. When do you need to intubate? 8. How is CHF management different in a hypotensive patient? 9. How would you approach an intubation like this on the ward? What are options to buy time until help is available? Why is this airway so tenuous?	
Key Moments	
Recognition of acute respiratory distress and need to intervene.	
Recognition of further deterioration and need to intubate.	

